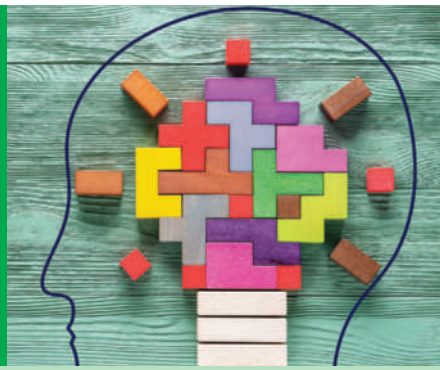
01

An Introduction to Information Systems



Principles	Learning Objectives
The value of information is directly linked to how it helps decision makers achieve organizational goals.	■ Discuss why it is important to study and understand information systems. ■ Describe the characteristics used to evaluate the quality of data.
Computers and information systems are constantly making it possible for organizations to improve the way they conduct business.	■ Name the components of an information system and describe several system characteristics.
Knowing the potential impact of information systems and having the ability to put this knowledge to work can result in a successful personal career, organizations that reach their goals and a society with a higher quality of life.	■ Identify the basic types of business information systems and discuss who uses them, how they are used and what kinds of benefits they deliver.
System users, business managers and information systems professionals must work together to build a successful information system.	■ Identify the major steps of the systems development process and state the goal of each.
Information systems must be applied thoughtfully and carefully so that society, business and industry can reap their enormous benefits.	■ Describe some of the threats to security and privacy that information systems and the Internet can pose. ■ Discuss the expanding role and benefits of information systems in business and industry.

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Why Learn About Information Systems?

Information systems are used in almost every imaginable profession. Sales representatives use information systems to advertise products, communicate with customers and analyze sales trends. Managers use them to make major decisions, such as whether to build a manufacturing plant or research a cancer drug. From a small music store to huge multinational companies, businesses of all sizes could not survive without information systems to perform accounting and finance operations. Regardless of your chosen career, you will use information systems to help you achieve goals.

This chapter presents an overview of information systems. The sections on hardware, software, databases, telecommunications, e-commerce and m-commerce, transaction processing and enterprise resource planning, information and decision support, special purpose systems, systems development, and ethical and societal issues are expanded to full chapters in the rest of the book. We will start by exploring the basics of information systems.

1.1 What is an Information System?

People and organizations use information every day. Many retail chains, for example, collect data from their shops to help them stock what customers want and to reduce costs. Businesses use information systems to increase revenues and reduce costs. We use automated teller machines outside banks and access information over the Internet. Information systems usually involve computers, and together they are constantly changing the way organizations conduct business. Today we live in an information economy. Information itself has value, and commerce often involves the exchange of information rather than tangible goods. Systems based on computers are increasingly being used to create, store and transfer information. Using information systems, investors make multimillion-euro decisions, financial institutions transfer billions of euros around the world electronically, and manufacturers order supplies and distribute goods faster than ever before. Computers and information systems will continue to change businesses and the way we live. To define an information system, we will start by examining what a system is.

What is a System?

system A set of elements or components that interact to accomplish goals.

A central concept of this book is that of a **system**. A system is a set of elements or components that interact to accomplish goals. The elements themselves and the relationships between them determine how the system works. Systems have inputs, processing mechanisms, outputs and feedback (see Figure 1.1). A system processes the input to create the output. For example, consider an automatic car wash. Tangible inputs for the process are a dirty car, water and various cleaning ingredients. Time, energy, skill and knowledge also serve as inputs to the system because they are needed to operate it.

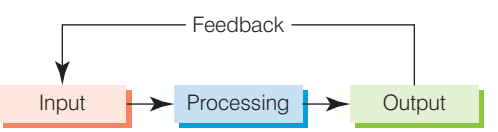


Figure 1.1 Components of a System *A system's four components consist of input, processing, output and feedback.*

The processing mechanisms consist of first selecting which cleaning option you want (wash only, wash with wax, wash with wax and hand dry, etc.) and communicating that to the operator of the car wash. Liquid sprayers shoot clean water, liquid soap or car wax depending on where your car is in the process and which options you selected. The output is a clean car. As in all systems,

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independent elements or components (the liquid sprayer, foaming brush and air dryer) interact to create a clean car. A feedback mechanism is your assessment of how clean the car is.

System performance can be measured in various ways. **Efficiency** is a measure of what is produced divided by what is consumed. For example, the efficiency of a motor is the energy produced (in terms of work done) divided by the energy consumed (in terms of electricity or fuel). Some motors have an efficiency of 50 per cent or less because of the energy lost to friction and heat generation.

efficiency A measure of what is produced divided by what is consumed.

Effectiveness is a measure of the extent to which a system achieves its goals. It can be computed by dividing the goals actually achieved by the total of the stated goals. For example, a company might want to achieve a net profit of €100 million for the year with a new information system. Actual profits, however, might only be €85 million for the year. In this case, the effectiveness is 85 per cent (85/100 = 85 per cent).

effectiveness A measure of the extent to which a system achieves its goals; it can be computed by dividing the goals actually achieved by the total of the stated goals.

Evaluating system performance also calls for using performance standards. A **system performance standard** is a specific objective of the system. For example, a system performance standard for a marketing campaign might be to have each sales representative sell €100,000 of a certain type of product each year (see Figure 1.2a). A system performance standard for a manufacturing process might be to produce no more than 1 per cent defective parts (see Figure 1.2b). After standards are established, system performance is measured and compared with the standard. Variances from the standard are determinants of system performance.

system performance standard A specific objective of the system.

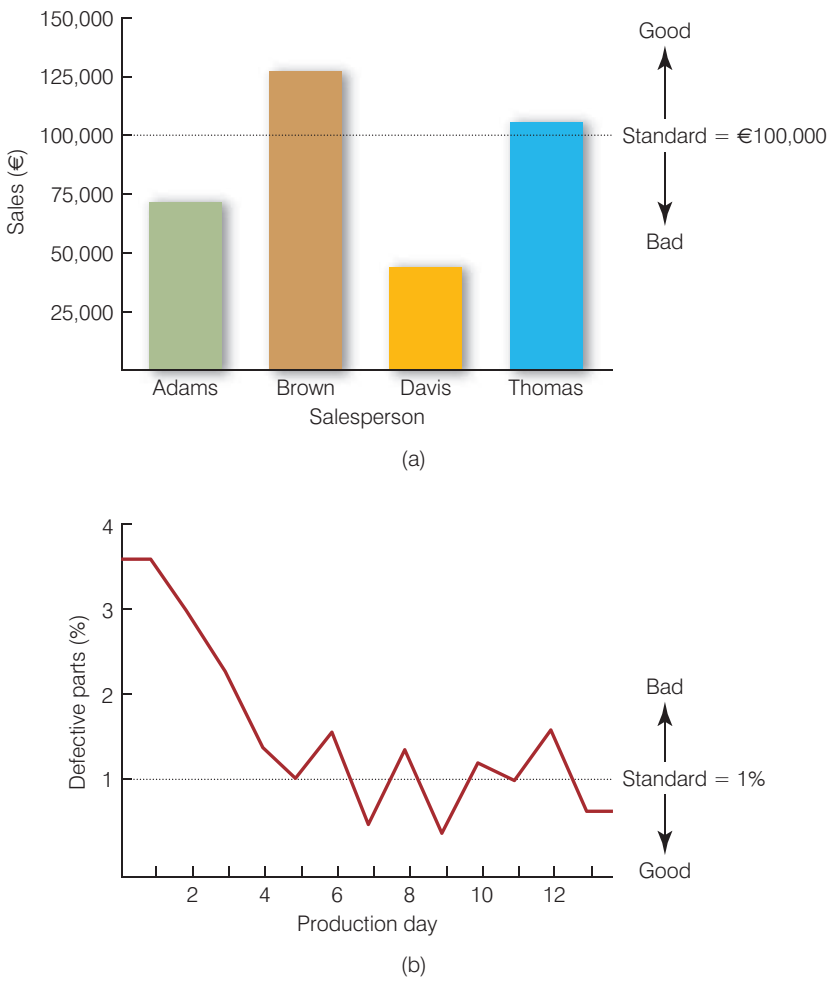


Figure 1.2 System Performance Standards
(a) Sales broken down by salesperson.
(b) Percentage of defective parts.

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What is Information?

Information is one of those concepts that we all seem intuitively able to grasp but find tricky to define. In the 1940s, mathematician Claude Shannon defined it as: information is that which reduces uncertainty. Shannon was working on the technical problems involved in sending messages over communication networks, and his concept of information is actually quite different from what we in business information systems mean by 'information'. Nevertheless, we can use his definition as a starting point. Imagine you are unsure of what today's weather will be like. Getting out of bed you open the curtains to see that the sun is shining. You now know a bit more about what it's going to be like: your uncertainty about the weather has been reduced. Therefore looking out of the window gave you information. When you turn on your radio and hear a detailed weather report, your uncertainty has been reduced further. When you look at the temperature gauge in your car, again your uncertainty has gone down. According to Shannon's definition, each of these events has therefore given you information.

However, his definition does not really capture what we would think of when we consider the information in, say, a management report. Therefore we simply define information as a collection of facts. These facts can take many forms. The temperature gauge in the car gives information in the form of a number. The radio gives audio information. Looking out of the window gives visual information. Other forms of information include text, images and video clips.

Another term that is closely related to information is 'data'. It's not intuitive but a philosopher might define data as 'variation'. To explain this: a blank page contains no data, but as soon as there is a mark on the page, that is, as soon as there is variation in the blankness, then data exist. Again this doesn't really capture what we mean by data in the context of business information systems. The traditional information systems view is that the input to an information system is data, and the output from the system is information. This means therefore that the difference between them is to do with how much processing has been done: unprocessed facts are data; processed facts are information. Unfortunately, however, this distinction is of little practical use. Therefore we will simply use the terms 'information' and 'data' interchangeably and define them as a collection of facts which can come in a variety of formats. (Incidentally, strictly speaking, the term data is plural, so we would say 'data are used' rather than 'data is used'. However, this is often not adhered to and we won't worry too much about it here.)

What is an Information System?

information system (IS) A set of interrelated components that collect, manipulate, store and disseminate information and provide a feedback mechanism to meet an objective.

input The activity of gathering and capturing data.

processing Converting or transforming input into useful outputs.

Now that we have defined the terms 'system' and 'information', we can define an information system: an **information system (IS)** is a set of interrelated components that collect, manipulate, store and disseminate information and provide a feedback mechanism to meet an objective. It is the feedback mechanism that helps organizations achieve their goals, such as increasing profits or improving customer service.

In information systems, **input** is the activity of gathering and capturing data. In producing employment payment, for example, the number of hours every employee works must be collected before the cheques can be calculated or printed. In a university grading system, instructors must submit student grades before a summary of grades for the semester can be compiled and sent to the students.

Processing means converting or transforming this input into useful outputs. Processing can involve making calculations, comparing data and taking alternative actions, and storing data for future use. In a payroll application, the number of hours each employee worked must be converted into net, or take-home, pay. Other inputs often include employee ID number and department. The required processing can first involve multiplying the number of hours

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worked by the employee's hourly pay rate to get gross pay. If weekly hours worked exceed basic hours, overtime pay might also be included. Then tax must be deducted along with contributions to health and life insurance or savings plans to get net pay.

After these calculations and comparisons are performed, the results are typically stored. Storage involves keeping data and information available for future use, including output.

Output involves producing useful information, usually in the form of documents and reports. Outputs can include paycheques for employees, reports for managers, and information supplied to stockholders, banks, government agencies and other groups. In addition, output from one system can become input for another. For example, output from a system that processes sales orders can be used as input to a customer billing system. Computers typically produce output on printers and display screens. Output can also be handwritten or manually produced reports, although this is not common.

output Production of useful information, often in the form of documents and reports.

Lastly, **feedback** is information from the system that is used to make changes to input or processing activities. For example, errors or problems might make it necessary to correct input data or change a process. Consider a payroll example. Perhaps the number of hours an employee worked was entered as 400 instead of 40 hours. Fortunately, most information systems check to make sure that data falls within certain ranges. For number of hours worked, the range might be from 0 to 100 hours because it is unlikely that an employee would work more than 100 hours in a week. The information system would determine that 400 hours is out of range and provide feedback. The feedback is used to check and correct the input on the number of hours worked to 40.

feedback Output that is used to make changes to input or processing activities.

Feedback is also important for managers and decision makers. For example, a furniture maker could use a computerized feedback system to link its suppliers and manufacturing plants. The output from an information system might indicate that inventory levels for mahogany and oak are getting low – a potential problem. A manager could use this feedback to decide to order more wood from a supplier. These new inventory orders then become input to the system. In addition to this reactive approach, a computer system can also be proactive – predicting future events to avoid problems. This concept, often called **forecasting**, can be used to estimate future sales and order more inventory before a shortage occurs. Forecasting is also used to predict the strength of hurricanes and where they will reach land, future stock-market values and who will win a political election.

forecasting Predicting future events.

The Characteristics of Valuable Information

To be valuable to managers and decision makers, information should have some and possibly all of the characteristics described in Table 1.1. Many shipping companies, for example, can determine the exact location of inventory items and packages in their systems, and this information makes them responsive to their customers. In contrast, if an organization's information is not accurate or complete, people can make poor decisions costing thousands, or even millions, of euros. Many claim, for example, that the collapse and bankruptcy of some companies, such as drug companies and energy-trading firms, was a result of inaccurate accounting and reporting information, which led investors and employees alike to misjudge the actual state of these companies' finances and suffer huge personal losses. As another example, if an inaccurate forecast of future demand indicates that sales will be very high when the opposite is true, an organization can invest millions of euros in a new plant that is not needed. Furthermore, if information is not relevant, not delivered to decision makers in a timely fashion, or too complex to understand, it can be of little value to the organization.

The value of information is directly linked to how it helps decision makers achieve their organization's goals. For example, the value of information might be measured in the time required to make a decision or in increased profits to the company. Consider a market forecast

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that predicts a high demand for a new product. If you use this information to develop the new product and your company makes an additional profit of €10,000, the value of this information to the company is €10,000 minus the cost of the information.

Table 1.1 Characteristics of Valuable Information

Characteristics	Definitions
Accessible	Information should be easily accessible by authorized users so they can obtain it in the right format and at the right time to meet their needs
Accurate	Accurate information is error free. In some cases, inaccurate information is generated because inaccurate data is fed into the transformation process
Complete	Complete information contains all the important facts, but not more facts than are necessary (see the Simple characteristic below)
Economical	Information should also be relatively economical to produce. Decision makers must always balance the value of information with the cost of producing it
Flexible	Flexible information can be used for a variety of purposes. For example, information on how much inventory is on hand for a particular part can be used by a sales representative in closing a sale, by a production manager to determine whether more inventory is needed, and by a financial executive to determine the total value the company has invested in inventory
Relevant	Relevant information is important to the decision maker
Reliable	Reliable information can be depended on. In many cases, the reliability of the information depends on the reliability of the data-collection method. In other instances, reliability depends on the source of the information. A rumour from an unknown source that oil prices might go up soon may not be reliable (even though it might be useful)
Secure	Information should be secure from access by unauthorized users
Simple	Information should be simple, not overly complex. Sophisticated and detailed information might not be needed. In fact, too much information can cause information overload, whereby a decision maker is unable to determine what is really important
Timely	Timely information is delivered when it is needed. Knowing last week's weather conditions will not help when trying to decide what coat to wear today
Verifiable	Information should be verifiable. This means that you can check it to make sure it is correct, perhaps by checking many sources for the same information

Manual and Computerized Information Systems

An information system can be manual or computerized. For example, some investment analysts manually draw charts and trend lines to assist them in making investment decisions. Tracking data on stock prices (input) over the last few months or years, these analysts develop patterns in graphical form (processing) that help them determine what stock prices are likely to do in the next few days or weeks (output). Some investors have made millions of euros using manual stock analysis information systems. Of course, today, many excellent computerized information systems

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follow stock indexes and markets and suggest when large blocks of stocks should be purchased or sold to take advantage of market discrepancies.

The components of a **computer-based information system (CBIS)** are illustrated in Figure 1.3. Information technology (IT) refers to hardware, software, databases and telecommunications. A business's **technology infrastructure** includes all the hardware, software, databases, telecommunications, people and procedures that are configured to collect, manipulate, store and process data into information. The technology infrastructure is a set of shared IS resources that form the foundation of each computer-based information system.

computer-based information system (CBIS) A single set of hardware, software, databases, telecommunications, people and procedures that is configured to collect, manipulate, store and process data into information.

technology infrastructure All the hardware, software, databases, telecommunications, people and procedures that are configured to collect, manipulate, store and process data into information.

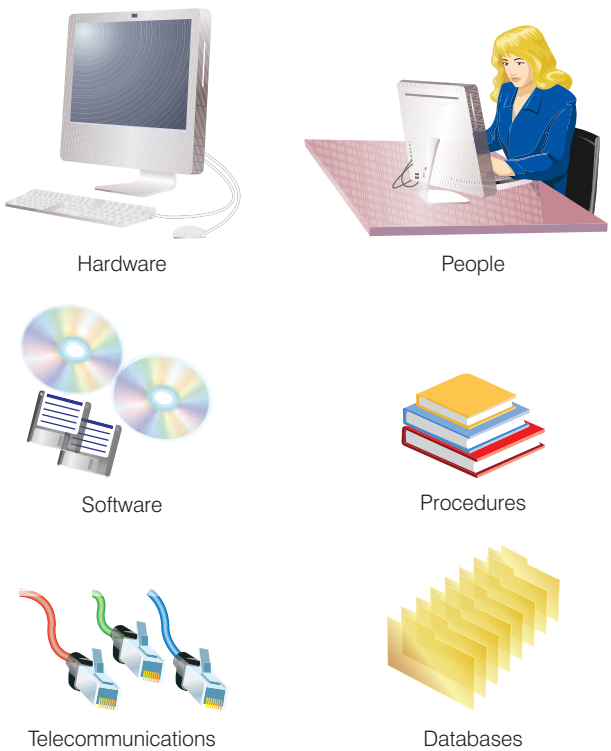


Figure 1.3 The Components of a Computer-Based Information System

Hardware

Hardware consists of computer equipment used to perform input, processing and output activities. Input devices include keyboards, mice and other pointing devices, automatic scanning devices and equipment that can read magnetic ink characters. Investment firms often use voice-response technology to allow customers to access their balances and other information with spoken commands. Processing devices include computer chips that contain the central processing unit and main memory. One processor chip, called the 'Bunny Chip' by some, mimics living organisms and can be used by the drug industry to test drugs instead of using animals, such as rats or bunnies.[1] The experimental chip could save millions of euros and months of time in drug research costs, as well as having a positive impact by reducing animal testing. Speed is an important part of assessing hardware. The TOP500 project (www.top500.org) has collected statistics on the world's fastest computers since 1993. Currently the fastest is the Summit computer at the Oak Ridge National Laboratory in Tennessee, USA. Summit is providing

hardware Any machinery (most of which uses digital circuits) that assists in the input, processing, storage and output activities of an information system.

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scientists with computing power to research problems in energy, artificial intelligence and human health. It is able to perform up to 143.5 petaflops; a petaflop is one thousand million million calculations per second.[2,3]

The many types of output devices include printers and computer screens. Bond traders, for example, often use an array of six or more computer screens to monitor bond prices and make split-second trades throughout each day. Another type of output device is a printer to print photos from a digital camera. Such printers accept the memory card direct from the camera. There are also many special-purpose hardware devices. Computerized event data recorders (EDRs) are now being placed into vehicles. Like an aeroplane black box, EDRs record a vehicle's speed, possible engine problems, a driver's performance and more. The technology is being used to monitor vehicle operation, determine the cause of accidents and investigate whether truck drivers are taking required breaks.

Software

software The computer programs that govern the operation of the computer.

Software consists of the computer programs that govern the operation of the computer. These programs allow a computer to process payroll, send bills to customers and provide managers with information to increase profits, reduce costs and provide better customer service. With software, people can work anytime at any place. Software, along with manufacturing tools, for example, can be used to fabricate parts almost anywhere in the world.[4] Software called 'Fab Lab' controls tools, such as cutters, milling machines and other devices. A Fab Lab system, which costs about €15,000, has been used to make radio frequency tags to track animals in Norway, engine parts to allow tractors to run on processed castor beans in India and many other fabrication applications.

The two types of software are system software, such as Microsoft Windows, which controls basic computer operations, including start-up and printing; and applications software, such as Microsoft Office, which allows you to accomplish specific tasks, including word processing and drawing charts. Sophisticated application software, such as Adobe Creative Suite, can be used to design, develop, print and place professional-quality advertising, brochures, posters, prints and videos on the Internet.

Databases

database An organized collection of electronic information.

A **database** is an organized collection of facts and information, typically consisting of two or more related data files. An organization's database can contain information on customers, employees, inventory, competitors' sales, online purchases and much more. Most managers and executives consider a database to be one of the most valuable parts of a computer-based information system. One California real estate development company uses databases to search for homes that are undervalued and purchase them at bargain prices.[5] It uses the database to analyze crime statistics, prices, local weather reports, school districts and more to find homes whose values are likely to increase. The database has helped the company realize an average 50 per cent return on investment. Increasingly, organizations are placing important databases on the Internet, which makes them accessible to many, including unauthorized users.

Telecommunications, Networks and the Internet

telecommunications The electronic transmission of signals for communications; enables organizations to carry out their processes and tasks through effective computer networks.

networks Computers and equipment that are connected in a building, around the country or around the world to enable electronic communications.

Telecommunications is the electronic transmission of signals for communications, which enables organizations to carry out their processes and tasks through computer networks. Large restaurant chains, for example, can use telecommunications systems and satellites to link hundreds of restaurants to plants and headquarters to speed credit card authorization and report sales and payroll data. **Networks** connect computers and equipment in a building, around the country or around the world to enable electronic communication. Investment firms can use wireless networks to connect thousands of investors with brokers or traders. Many hotels use wireless telecommunications to allow guests to

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connect to the Internet, retrieve voice messages and exchange email without plugging their computers or mobile devices into a phone socket. Wireless transmission also allows drones, such as Boeing's Scan Eagle, to fly using a remote control system and monitor buildings and other areas.

The **Internet** is the world's largest computer network, actually consisting of thousands of interconnected networks, all freely exchanging information. Research firms, colleges, universities, schools and businesses are just a few examples of organizations using the Internet. People use the Internet to research information, buy and sell products and services, make travel arrangements, conduct banking, and download music and videos, among other activities. After downloading music, you can use audio software to change a song's tempo, create mixes of your favourite tunes and modify sound tracks to suit your personal taste. You can even mix two or more songs simultaneously, which is called 'mashing'. You can also use many of today's mobile phones to connect to the Internet from around the world and at high speeds.[6] This not only speeds communications but allows you to conduct business electronically. Some airline companies are providing Internet service on their flights so that travellers can send and receive email, check investments and browse the Internet. Internet users can create blogs (weblogs) to store and share their thoughts and ideas with others around the world.[7] You can also record and store TV programmes on computers or special viewing devices and watch them later.[8] Often called 'place shifting', this technology allows you to record TV programmes at home and watch them at a different place when it's convenient.

Internet The world's largest computer network, actually consisting of thousands of interconnected networks, all freely exchanging information.

The World Wide Web (WWW), or the web, is a network of links on the Internet to documents containing text, graphics, video and sound. Information about the documents and access to them are controlled and provided by tens of thousands of special computers called 'web servers'. The web is one of many services available over the Internet and provides access to many hundreds of millions of documents. Widely available Internet access has allowed the development of **cloud computing**, where software and data storage are provided as an Internet service and are accessed via a web browser.

cloud computing A computing environment where software and storage are provided as an Internet service and are accessed via a web browser.

The technology used to create the Internet is also being applied within companies and organizations to create **intranets**, which allow people within an organization to exchange information and work on projects. One company, for example, uses an intranet to connect its 200 global operating companies and 20,000 employees. An **extranet** is a network based on web technologies that allows selected outsiders, such as business partners and customers, to access authorized resources of a company's intranet. Companies can move all or most of their business activities to an extranet site for corporate customers. Many people use extranets every day without realizing it – to track shipped goods, order products from their suppliers or access customer assistance from other companies. If you log on to the FedEx site (www.fedex.com) to check the status of a package, for example, you are using an extranet.

intranets An internal company network built using Internet and World Wide Web standards and products that allows people within an organization to exchange information and work on projects.

extranet A network based on web technologies that allows selected outsiders, such as business partners, suppliers or customers, to access authorized resources of a company's intranet.

People

People are the most important element in most computer-based information systems. The people involved include users of the system and information systems personnel, including all the people who manage, run, program and maintain the system.

Procedures

Procedures include the strategies, policies, methods and rules for using the CBIS, including the operation, maintenance and security of the computer. For example, some procedures describe when each program should be run. Others describe who can access facts in the database, or what to do if a disaster, such as a fire, earthquake or hurricane, renders the CBIS unusable. Good procedures can help companies take advantage of new opportunities and avoid potential disasters. Poorly developed and inadequately implemented procedures, however, can cause people to waste their time on useless rules or result in inadequate responses to disasters, such as hurricanes or tornadoes.

procedures The strategies, policies, methods and rules for using a CBIS.

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1.2 Business Information Systems

The most common types of information systems used in business organizations are those designed for electronic and mobile commerce, transaction processing, management information and decision support. In addition, some organizations employ special-purpose systems, such as virtual reality, that not every organization uses. Together, these systems help employees in organizations accomplish routine and special tasks – from recording sales, processing payrolls and supporting decisions in various departments, to examining alternatives for large-scale projects and opportunities. Often in large organizations one information system is used to accomplish all of these tasks. In others, separate systems are used. When one system is used it is called an Enterprise Resource Planning System, and it does most if not all of the tasks of the other systems shown in Figure 1.4.

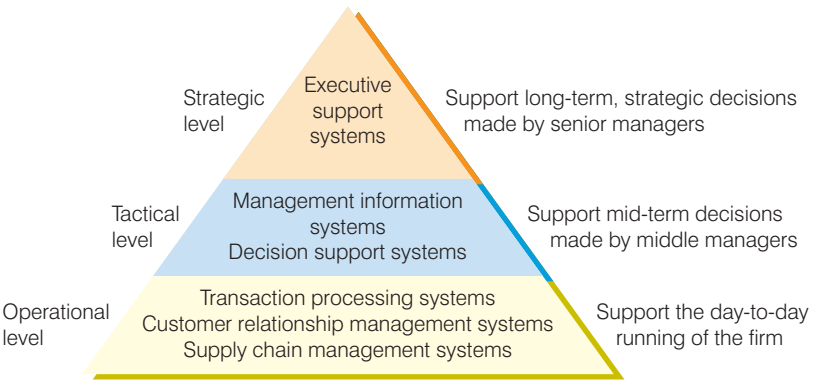


Figure 1.4 Business Information Systems
The triangle shows the main types of information system used in businesses and the level within the business where they tend to be used.

Information Systems @ Work



Translating Shakespeare into Xhosa

Xhosa is an official language in South Africa and Zimbabwe, spoken by Bantu people. It uses three click-consonant sounds for X, Q and C. So the name Xhosa is pronounced by making a clicking or TCHICK sound for the X, resulting in TCHICK-hosa. A way of writing Xhosa using the Latin alphabet was developed in the early 19th century and the language is available as an option on Google Translate.

Many people using Google Translate for the first time are amazed by its abilities. If you haven't tried it, visit your local site (it's www.translate.google.co.uk in the UK), enter some text, choose a language and hit translate. If you actually speak the language you translated into, how did it do? Does the translation make sense and does it say what you meant it to say? If you don't speak that language, copy the translated text and translate it back into the first language – it probably won't be exactly the same, but does it still mean roughly what you said in the first place? Try translating the text through several other languages – English to German to French to Chinese and then back to English – what does it say now? This game is actually popular fodder for YouTube videos – YouTubers translate the lyrics of a popular song into another language, then back into the first language, and record themselves singing exactly what they ended up with. However, this technology has more serious applications.

So how does it work? It relies on the one thing that Google is extremely good at – processing lots and lots of data, very, very quickly. Sorry to disappoint you if you thought the computer could actually understand what you are saying, but in fact it's just doing some statistics. There are many

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books that have been translated by a human from one language into another, say Victor Hugo's *Notre Dame de Paris*, which was written in French. To translate a French phrase, all the computer has to do is find the phrase in *Notre Dame de Paris*, or some other French book that has been translated, then go to the corresponding page and line in the translation, and return that to the user. But how does the computer know that the page and line are the exact words that translate the exact phrase that the user wrote? That's where the statistics and data processing come in. The computer doesn't just search for the phrase in one book, it looks in say 100 books (the actual figure Google uses is difficult to track down and probably varies) and finds roughly where the phrase will be in each. This will give 100 samples of text that contain the translation somewhere within them. It then chooses the phrase that appears most often in the sample of 100.

For example, the line 'igubu lokukhaphela ingoma endiyivuma ngesingqala' in a Xhosa poem by Ncumisa Garishe MaMiya gets translated as 'the drum of song that I sing in a thunderstorm'. This sequence of words probably only appears in that one poem and nowhere else, so to translate it Google might look for three phrases 'the drum of song', 'that I sing' and 'in a thunderstorm' in its source material to find a translation for each. Then it will put all three together, perhaps also applying some rules it has been taught about grammar.

Taking just the word 'ingoma', which means 'song', if the computer finds 100 sources where this word appears, and in 97 of the translations at around about the right point in the text the word 'song' appears, then that would seem to be a correct translation with 97 per cent confidence. The computer doesn't have to understand anything about language to do this. Given enough sample text, the computer could translate all of Shakespeare. (Trying this, you get 'ngoku kubusika kokunganeliseki kwethu' for 'now is the winter of our discontent'.)

Google Translate was developed under the leadership of computer scientist Franz Och and originally used documents created by the United Nations. The UN has a need to create a lot of written material accurately in a number of languages, and all of this material is freely available, which made it a good source to use. (We should point out that Google did not invent statistical translation.)

Questions

1. Try the exercises suggested in the case – how did Google Translate do?
2. Could this technology be used by a global firm or is the need for absolute accuracy so great that a human translator would always be needed?
3. What are some of the problems with using Google Translate to translate a web page?
4. What would the computer need in order to translate 'Shakespeare into Xhosa'?

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Enterprise Systems: Transaction Processing Systems and Enterprise Resource Planning

Enterprise systems help organizations perform and integrate important tasks, such as paying employees and suppliers, controlling inventory, sending out invoices and ordering supplies. In the past, companies accomplished these tasks using traditional transaction processing systems. Today, they are increasingly being performed by enterprise resource planning systems. For example, Whirlpool Corporation, the large appliance maker, used enterprise resource planning to reduce inventory levels by 20 per cent and cut about 5 per cent from its freight and warehousing

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costs by providing managers with information about inventory levels and costs.[9] The new system may have also helped the company increase its revenues by about €0.7 billion.

Transaction Processing Systems

Since the 1950s, computers have been used to perform common business applications. Many of these early systems were designed to reduce costs by automating routine, labour-intensive business transactions. A **transaction** is any business-related exchange, such as payments to employees, sales to customers or payments to suppliers. Thus, processing business transactions was the first computer application developed for most organizations. A **transaction processing system (TPS)** is an organized collection of people, procedures, software, databases and devices used to record completed business transactions. If you understand a transaction processing system, you understand basic business operations and functions.

transaction Any business-related exchange, such as payments to employees, sales to customers and payments to suppliers.

transaction processing system (TPS) An organized collection of people, procedures, software, databases and devices used to record completed business transactions.

One of the first business systems to be computerized was the payroll system. The primary inputs for a payroll TPS are the number of employee hours worked during the week and the pay rate. The primary output consists of paycheques. Early payroll systems produced employee paycheques and related reports required by tax authorities. Other routine applications include sales ordering, customer billing and customer relationship management, and inventory control. Some car companies, for example, use their TPSs to buy billions of euros of needed parts each year through websites. Because these systems handle and process daily business exchanges or transactions, they are all classified as TPSs.

Enterprise Resource Planning

An **enterprise resource planning (ERP) system** is a set of integrated programs that manages the vital business operations for an entire multisite, global organization. An ERP system can replace many applications with one unified set of programs, making the system easier to use and more effective.

enterprise resource planning (ERP) system A set of integrated programs capable of managing a company's vital business operations for an entire multisite, global organization.

Although the scope of an ERP system might vary from company to company, most ERP systems provide integrated software to support manufacturing and finance. In such an environment, a forecast is prepared that estimates customer demand for several weeks. The ERP system checks what is already available in finished product inventory to meet the projected demand. Manufacturing must then produce inventory to eliminate any shortfalls. In developing the production schedule, the ERP system checks the raw material and packing material inventories and determines what needs to be ordered to meet the schedule. Most ERP systems also have a purchasing subsystem that orders the needed items. In addition to these core business processes, some ERP systems can support functions such as human resources, sales and distribution. The primary benefits of implementing an ERP system include easing adoption of improved work processes and increasing access to timely data for decision making.

An important type of transaction processing system handles transactions made electronically over the web. **E-commerce** involves any business transaction executed electronically between companies (business-to-business, 'B2B'), companies and consumers (business-to-consumer, 'B2C'), consumers and other consumers (consumer-to-consumer, 'C2C'), business and the public sector, and consumers and the public sector. You might assume that e-commerce is reserved mainly for consumers visiting websites for online shopping, but web shopping is only a small part of the e-commerce picture; the major volume of e-commerce – and its fastest growing segment – is business-to-business (B2B) transactions that make purchasing easier for corporations. This growth is being stimulated by increased Internet access, growing user confidence, better payment systems and rapidly improving Internet and web security. E-commerce also offers opportunities for small businesses to market and sell at a low cost worldwide, allowing them to enter the global

e-commerce Any business transaction executed electronically between companies (business-to-business), companies and consumers (business-to-consumer), consumers and other consumers (consumer-to-consumer), business and the public sector, and consumers and the public sector.

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market. **Mobile commerce (m-commerce)** refers to transactions conducted anywhere, anytime. M-commerce relies on wireless communications that managers and corporations use to place orders and conduct business with handheld computers, portable phones, laptop computers connected to a network and other mobile devices (see Figure 1.5, which shows a mobile phone being used to make a contactless payment).

mobile commerce (m-commerce) Conducting business transactions electronically using mobile devices such as smartphones.



Figure 1.5 Contactless Payment *M-commerce means that it is possible to pay for items using a smartphone or other mobile device.*

E-commerce offers many advantages for streamlining work activities. Figure 1.6 provides a brief example of how e-commerce can simplify the process of purchasing new office furniture from an office-supply company. In the manual system, a corporate office worker must get approval for a purchase that exceeds a certain amount. That request goes to the purchasing department, which generates a formal purchase order to procure the goods from the approved vendor. Business-to-business e-commerce automates the entire process. Employees go directly to the supplier's website, find the item in a catalogue and order what they need at a price set by their company. If approval is required, the approver is notified automatically. As the use of e-commerce systems grows, companies are phasing out their traditional systems. The resulting growth of e-commerce is creating many new business opportunities.

E-commerce can enhance a company's stock prices and market value. Today, several e-commerce firms have teamed up with more traditional brick-and-mortar businesses to draw from each other's strengths. For example, e-commerce customers can order products on a website and pick them up at a nearby store.

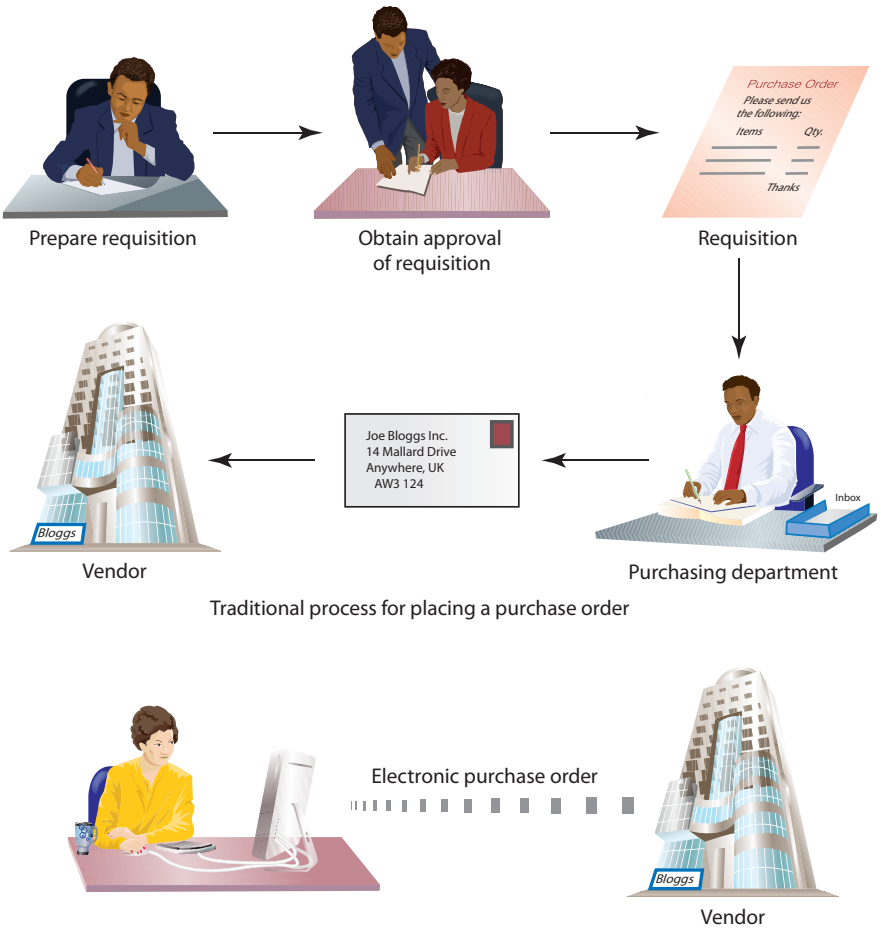
In addition to e-commerce, business information systems use telecommunications and the Internet to perform many related tasks. Electronic procurement (e-procurement), for example, involves using information systems and the Internet to acquire parts and supplies.

Electronic business (e-business) goes beyond e-commerce and e-procurement by using information systems and the Internet to perform all business-related tasks and functions, such as accounting, finance, marketing, manufacturing and human resource activities. E-business also includes working with customers, suppliers, strategic partners and stakeholders. Compared with traditional business strategy, e-business strategy is flexible and adaptable.

electronic business (e-business) Using information systems and the Internet to perform all business-related tasks and functions.

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Figure 1.6
E-Commerce Greatly Simplifies Purchasing



MIS and DSS

The benefits provided by an effective TPS are tangible and justify their associated costs in computing equipment, computer programs, and specialized personnel and supplies. A TPS can speed business activities and reduce clerical costs. Although early accounting and financial TPSs were already valuable, companies soon realized that they could use the data stored in these systems to help managers make better decisions, whether in human resource management, marketing or administration. Satisfying the needs of managers and decision makers continues to be a major factor in developing information systems.

Management Information Systems

management information system (MIS) An organized collection of people, procedures, software, databases and devices that provides routine information to managers and decision makers.

A **management information system (MIS)** is an organized collection of people, procedures, software, databases and devices that provides routine information to managers and decision makers. An MIS focuses on operational efficiency. Marketing, production, finance and other functional areas are supported by MISs and linked through a common database. MISs typically provide standard reports generated with data and information from the TPS, meaning the output of a TPS is the input to an MIS. Producing a report that describes inventory that should be ordered is an example of an MIS.

MISs were first developed in the 1960s and typically used information systems to produce managerial reports. In many cases, these early reports were produced periodically – daily,

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weekly, monthly or yearly. Because of their value to managers, MISs have proliferated throughout the management ranks. For instance, the total payroll summary report produced initially for an accounting manager might also be useful to a production manager to help monitor and control labour and job costs.

Decision Support Systems

By the 1980s, dramatic improvements in technology resulted in information systems that were less expensive but more powerful than earlier systems. People at all levels of organizations began using personal computers to do a variety of tasks; they were no longer solely dependent on the IS department for all their information needs. People quickly recognized that computer systems could support additional decision-making activities. A **decision support system (DSS)** is an organized collection of people, procedures, software, databases and devices that support problem-specific decision making. The focus of a DSS is on making effective decisions. Whereas an MIS helps an organization 'do things right', a DSS helps a manager 'do the right thing'.

decision support system (DSS) An organized collection of people, procedures, software, databases and devices used to support problem-specific decision making.

In addition to assisting in all aspects of problem-specific decision making, a DSS can support customers by rapidly responding to their phone and email enquiries. A DSS goes beyond a traditional MIS by providing immediate assistance in solving problems. Many of these problems are unique and complex, and information is often difficult to obtain. For instance, a car manufacturer might try to determine the layout for its new manufacturing facility. Traditional MISs are seldom used to solve these types of problems; a DSS can help by suggesting alternatives and assisting in final decision making.

DSSs are used when the problem is complex and the information needed to make the best decision is difficult to obtain and use. So a DSS also involves managerial judgement and perspective. Managers often play an active role in developing and implementing the DSS. A DSS recognizes that different managerial styles and decision types require different systems. For example, two production managers in the same position trying to solve the same problem might require different information and support. The overall emphasis is to support, rather than replace, managerial decision making.

The essential elements of a DSS include a collection of models used to support a decision maker or user (model base), a collection of facts and information to assist in decision making (database), and systems and procedures (dialogue manager or user interface) that help decision makers and other users interact with the DSS. Software is often used to manage the database – the database management system (DBMS) – and the model base – the model management system (MMS).

In addition to DSSs for managers, group decision support systems and executive support systems use the same approach to support groups and executives.[10] A group decision support system, also called a group support system, includes the DSS elements just described, and software, called groupware, to help groups make effective decisions. An executive support system, also called an executive information system, helps top-level managers, including a firm's CEO, divisional directors and members of the board of directors, make better decisions. An executive support system can assist with strategic planning, top-level organizing and staffing, strategic control and crisis management.

Knowledge Management, Artificial Intelligence, Expert Systems and Virtual Reality

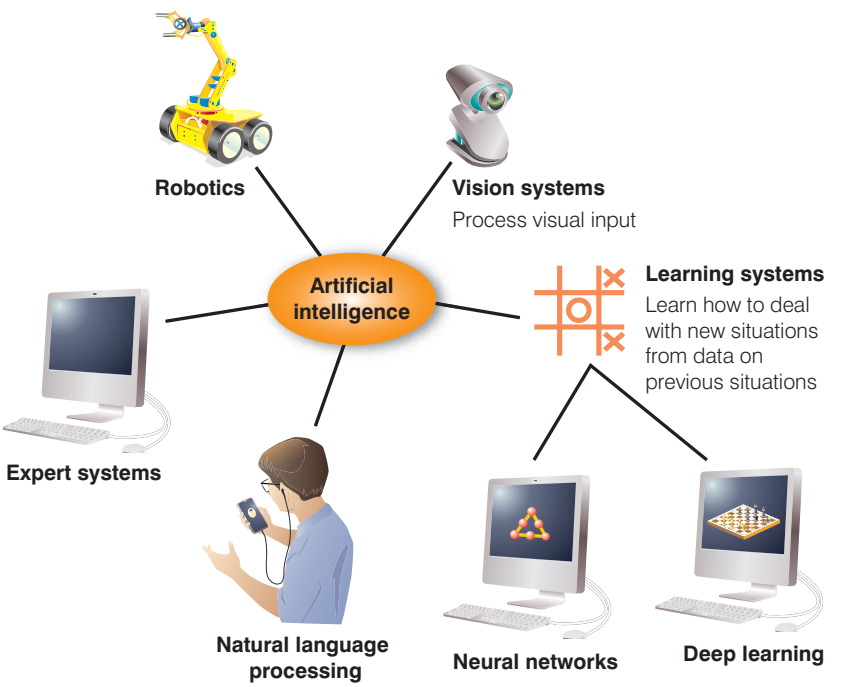
Also, organizations often rely on specialized systems. Many use knowledge management systems (KMSs), an organized collection of people, procedures, software, databases and devices to create, store, share and use the organization's knowledge and experience. According to a survey of CEOs, firms that use KMSs are more likely to innovate and perform better.[11]

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In addition to knowledge management, companies use other types of specialized systems. The Nissan Motor Company, for example, has developed a specialized system for its vehicles called 'Lane Departure Prevention' that nudges a car back into the correct lane if it veers off course.[12,13] The system uses cameras and computers to adjust braking to get the vehicle back on course. The system switches off when the driver uses turn signals to change lanes. Other specialized systems are based on the notion of **artificial intelligence (AI)**, in which the computer system takes on the characteristics of human intelligence. The field of artificial intelligence includes several subfields (see Figure 1.7). Computer scientists are predicting that deep learning will play a massive role in healthcare: computers will be used to analyze medical images, process health records and will fully automate certain teleoperated manipulation tasks of the surgical procedure (which means they will improve robot assisted surgery).[14]

artificial intelligence (AI) The ability of computer systems to mimic or duplicate the functions or characteristics of the human brain or intelligence.

Figure 1.7 The Major Elements of Artificial Intelligence



Learning Systems

Robotics is an area of AI in which machines take over complex, dangerous, routine or boring tasks, such as welding car frames or assembling computer systems and components. Vision systems allow robots and other devices to 'see', store and process visual images. Natural language processing involves computers understanding and acting on verbal or written commands in English, Spanish or other human languages. Learning systems allow computers to learn from past mistakes or experiences, such as playing games or making business decisions, and neural networks is a branch of AI that allows computers to recognize and act on patterns or trends. Some successful stock, options and futures traders use neural networks to spot trends and make them more profitable with their investments. State of the art AI is impressive. In 2011 the IBM supercomputer Watson competed against and beat two human champions in the game show *Jeopardy*. In December 2018, Google's Waymo company launched driverless taxis in four suburbs of Phoenix, Arizona in the USA. Riders use an app similar to the Uber app and, for the time being, the cars all have a human driver present to intervene in case of an emergency.[15]

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Expert Systems

Expert systems give the computer the ability to make suggestions and act like an expert in a particular field. It can help the novice user perform at the level of an expert. The unique value of expert systems is that they allow organizations to capture and use the wisdom of experts and specialists. Therefore, years of experience and specific skills are not completely lost when a human expert dies, retires or leaves for another job. Expert systems can be applied to almost any field or discipline. They have been used to monitor nuclear reactors, perform medical diagnoses, locate possible repair problems, design and configure IS components, perform credit evaluations and develop marketing plans for a new product or new investment strategy. The collection of data, rules, procedures and relationships that must be followed to achieve value or the proper outcome is contained in the expert system's **knowledge base**.

expert systems A system that gives a computer the ability to make suggestions and act like an expert in a particular field (or) hardware and software that stores knowledge and makes inferences, similar to a human expert.

knowledge base A component of an expert system that stores all relevant information, data, rules, cases and relationships used by the expert system.

Virtual Reality

Virtual reality is the simulation of a real or imagined environment that can be experienced visually in three dimensions. Originally, virtual reality referred to immersive virtual reality, which means the user becomes fully immersed in an artificial, computer-generated 3D world. The virtual world is presented in full scale and relates properly to the human size. It can represent any 3D setting, real or abstract, such as a building, an archaeological excavation site, the human anatomy, a sculpture or a crime scene reconstruction. Virtual worlds can be animated, interactive and shared. Through immersion, the user can gain a deeper understanding of the virtual world's behaviour and functionality. Virtual reality can also refer to applications that are not fully immersive, such as mouse-controlled navigation through a 3D environment on a graphics monitor, stereo viewing from the monitor via stereo glasses, stereo projection systems and others.

virtual reality The simulation of a real or imagined environment that can be experienced visually in three dimensions.

A variety of input devices, such as head-mounted displays, data gloves, joysticks and handheld wands, allow the user to navigate through a virtual environment and to interact with virtual objects. Directional sound, tactile and force feedback devices, voice recognition and other technologies enrich the immersive experience. Because several people can share and interact in the same environment, virtual reality can be a powerful medium for communication, entertainment and learning.

It is difficult to predict where information systems and technology will be in 10 to 20 years. It seems, however, that we are just beginning to discover the full range of their usefulness. Technology has been improving and expanding at an increasing rate; dramatic growth and change are expected for years to come. Without question, a knowledge of the effective use of information systems will be critical for managers both now and in the long term. But how are these information systems created?

1.3 Systems Development

Systems development is the activity of creating or modifying business systems. Systems development projects can range from small to very large in fields as diverse as stock analysis and video game development. People inside a company can develop systems or companies can use outsourcing, hiring an outside company to perform some or all of a systems development project. Outsourcing allows a company to focus on what it does best and delegate other functions to companies with expertise in systems development. Outsourcing, however, is not the best alternative for all companies. An alternative is agile systems development where systems are developed rapidly in close partnership with users. Working together, parts of the system are developed, tested, modified and refined over and over again until a usable system emerges.

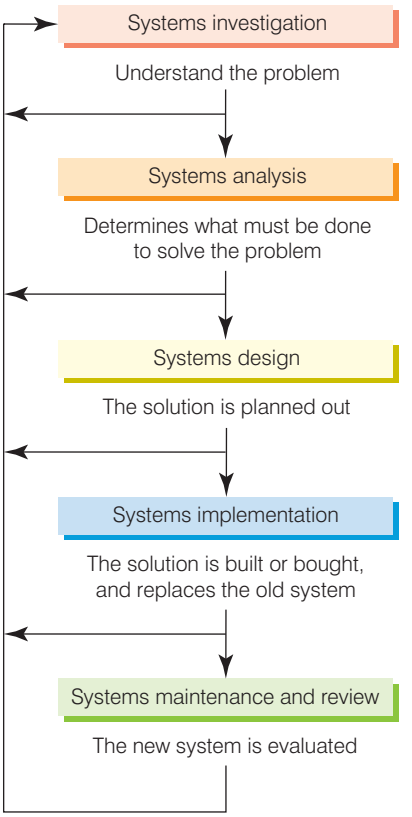
systems development The activity of creating or modifying existing business systems.

Developing information systems to meet business needs is highly complex and difficult – so much so that it is common for IS projects to overrun budgets and exceed scheduled

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completion dates. Her Majesty's Revenue and Customs (HMRC), which collects taxes in the UK, settled out of court with an outsourcing company to recover funds lost due to a tax-related mistake caused by a failed systems development project.[16] The failed project overpaid about €2.5 billion to some families with children or taxpayers in a low-income tax bracket. One strategy for improving the results of a systems development project is to divide it into several steps, each with a well-defined goal and set of tasks to accomplish (see Figure 1.8). These steps are summarized next.

Figure 1.8 An Overview of Systems Development



Systems Investigation and Analysis

The first two steps of systems development are systems investigation and analysis. The goal of the systems investigation is to gain a clear understanding of the problem to be solved or opportunity to be addressed. A cruise line company, for example, might launch a systems investigation to determine whether a development project is feasible to automate purchasing at ports around the world. After an organization understands the problem, the next question is, 'Is the problem worth solving?' Given that organizations have limited resources – people and money – this question deserves careful consideration. If the decision is to continue with the solution, the next step, systems analysis, defines the problems and opportunities of the existing system. During systems investigation and analysis, as well as design maintenance and review, discussed next, the project must have the complete support of top-level managers and focus on developing systems that achieve business goals.[17]

Systems Design, Implementation, and Maintenance and Review

Systems design determines how the new system will work to meet the business needs defined during systems analysis. Systems implementation involves creating or acquiring the various

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system components (hardware, software, databases, etc.) defined in the design step, assembling them and putting the new system into operation. The purpose of systems maintenance and review is to check and modify the system so that it continues to meet changing business needs.

1.4 Information Systems in Society, Business and Industry

Information systems have been developed to meet the needs of all types of organizations and people, and their use is spreading throughout the world to improve the lives and business activities of many citizens. To provide their enormous benefits, however, information systems must be implemented with thought and care. The speed and widespread use of information systems opens users to a variety of threats from unethical people.

Ethical and Societal Issues



Ethical Robots

'A robot may not injure a human being or, through inaction, allow a human being to come to harm.'

This is Isaac Asimov's famous first law of robotics and means that a robot must never harm a human being. But what if a robot had to choose between one of two actions, both of which would cause harm? Believe it or not, there are devices interacting with humans right now that at any time could be required to make such a decision. Driverless vehicles are a good example and are currently being tested and used on roads all over the world. What would happen if a child ran out in front of a driverless car and the car could swerve, but if it did swerve it would hit another child standing by the side of the road? According to *Nature* magazine, people often respond that the child at the side is more innocent than the one who ran out and therefore the robot should not swerve. But what if two children ran out and only one stood at the side? Or if it was an adult at the side? What if the adult looked strong and healthy and the robot calculated that they were more likely than the child to survive being hit by the car?

Answers to such questions often rely on what is known to philosophers as the doctrine of double effect: sometimes it is permissible to cause a harm as a side effect (or 'double effect') of bringing about a good result. Analyzing these situations is difficult for a decision-making program. The robot must predict two futures and assess whether the actions in each is not allowed because it causes harm, or permissible because the harm is only a side effect of causing good.

One question is whether this should be programmed into the robot or if the robot should be allowed to learn it for itself? Some computer scientists, like Michael Fisher from the University of Liverpool in the UK, think that rule-based systems could reassure the public. He told *Nature* magazine, 'People are going to be scared of robots if they're not sure what it's doing', he says. 'But if we can analyse and prove the reasons for their actions, we are more likely to surmount that trust issue.' On the other hand, the machine-learning approach (where the robot has to learn what is best for itself) would create robots that can learn from experience, which might ultimately make them more useful than a robot that must follow pre-programmed rules. According to *Nature*, many roboticists say that the best way forward will be a combination of the two approaches. The challenge is to make them work together.

Recently a series of researchers at MIT created the Moral Machine, an online platform to collect data from people all over the world about human opinions on how machines should make decisions when faced with moral dilemmas like these. The site presents participants with scenarios such as brake failure resulting in the deaths of two elderly pedestrians or four young passengers.

(continued)

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For each scenario, two possibilities are given and participants must choose which one is preferable. For its first major publication, the site was able to gather 40 million decisions from millions of people in over 230 countries and territories. Findings show that humans have a preference to spare humans over animals, spare more lives, and spare young lives, and that both male and female respondents have a preference for sparing female lives.

Questions

1. Should the robot swerve or not in each of the examples given? Can you justify your answer?
2. Would you be comfortable taking a robotic taxi? What about a driverless car?
3. Could a robot ever learn whether or not to swerve? Wouldn't it need to be in the situations many times in order to learn that?
4. View the Moral Machine at www.moralmachine.mit.edu and judge a few of the scenarios. Are there any where you are unable to make a choice?

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Security, Privacy and Ethical Issues in Information Systems and the Internet

Although information systems can provide enormous benefits, they do have some drawbacks.[18] Computer-related mistakes are also a concern. In Japan, a financial services firm had trading losses of ¥245 million due to a typing mistake in entering a trade. In another case, criminals stole carbon credits worth about £30 million from a European carbon credit market.[19]

Increasingly, the ethical use of systems has been highlighted in the news. Ethical issues concern what is generally considered right or wrong. Some IS professionals believe that computers may create new opportunities for unethical behaviour. For example, a faculty member of a medical school falsified computerized research results to get a promotion – and a higher salary. In another case, a company was charged with using a human resource information system to time employee layoffs and firings to avoid paying pensions. More and more, the Internet is also associated with unethical behaviour. Unethical investors have placed false rumours or incorrect information about

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a company on the Internet and tried to influence its stock price to make money. Information theft, such as stealing credit card numbers and other personal information, is another issue.

To protect against these threats, you can install security and control measures. For example, many software products can detect and remove viruses and spam, or unwanted email, from computer systems. Information systems can help reduce other types of crime as well. In a New Zealand city, a free computer centre has cut vandalism by keeping young people off the street and giving residents a sense of pride. When a pair of headphones disappeared from the centre, the community rallied to make sure that they were promptly returned.

You can install firewalls (software and hardware that protect a computer system or network from outside attacks) to avoid viruses and prevent unauthorized people from gaining access to your computer system. You can also use identification numbers and passwords. Some security experts propose installing web cameras and hiring 'citizen spotters' to monitor them. Use of information systems also raises work concerns, including job loss through increased efficiency and some potential health problems from making repetitive motions. Ergonomics, the study of designing and positioning workplace equipment, can help you avoid health-related problems of using computer systems.

Computer and Information Systems Literacy

In the twenty-first century, business survival and prosperity have continued to become more difficult. For example, increased mergers among former competitors to create global conglomerates, continued downsizing of corporations to focus on their core businesses and to improve efficiencies, efforts to reduce trade barriers, and the globalization of capital all point to the increased internationalization of business organizations and markets. In addition, business issues and decisions are becoming more complex and must be made faster. Whatever career path you take, understanding information systems will help you cope, adapt and prosper in this challenging environment.

A knowledge of information systems will help you make a significant contribution to the job. It will also help you advance in your chosen career or field. Managers are expected to identify opportunities to implement information systems to improve their business. They are also expected to lead IS projects in their areas of expertise. To meet these personal and organizational goals, you must acquire both computer literacy and information systems literacy. **Computer literacy** is a knowledge of computer systems and equipment and the ways they function. It stresses equipment and devices (hardware), programs and instructions (software), databases and telecommunications.

computer literacy Knowledge of computer systems and equipment and the ways they function; it stresses equipment and devices (hardware), programs and instructions (software), databases and telecommunications.

Information systems literacy goes beyond knowing the fundamentals of computer systems and equipment. **Information systems literacy** is the knowledge of how data and information are used by individuals, groups and organizations. It includes knowledge of computer technology and the broader range of information systems. Most importantly, however, it encompasses how and why this technology is applied in business. Knowing about various types of hardware and software is an example of computer literacy. Knowing how to use hardware and software to increase profits, cut costs, improve productivity and increase customer satisfaction is an example of information systems literacy. Information systems literacy can involve recognizing how and why people (managers, employees, stockholders and others) use information systems; being familiar with organizations, decision-making approaches, management levels and information needs; and understanding how organizations can use computers and information systems to achieve their goals. Knowing how to deploy transaction processing, management information, decision support and expert systems to help an organization achieve its goals is a key aspect of information systems literacy.

information systems literacy Knowledge of how data and information are used by individuals, groups and organizations.

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Information Systems in the Functional Areas of Business

Information systems are used in all functional areas and operating divisions of business. In finance and accounting, information systems forecast revenues and business activity, determine the best sources and uses of funds, manage cash and other financial resources, analyze investments and perform audits to make sure that the organization is financially sound and that all financial reports and documents are accurate. Sales and marketing use information systems to develop new goods and services (product analysis), select the best location for production and distribution facilities (place or site analysis), determine the best advertising and sales approaches (promotion analysis) and set product prices to get the highest total revenues (price analysis).

In manufacturing, information systems process customer orders, develop production schedules, control inventory levels and monitor product quality. In addition, information systems help to design products (computer-assisted design or CAD), manufacture items (computer-assisted manufacturing or CAM), and integrate machines or pieces of equipment (computer-integrated manufacturing or CIM). Human resource management uses information systems to screen applicants, administer performance tests to employees, monitor employee productivity and more. Legal information systems analyze product liability and warranties and help to develop important legal documents and reports.

Information Systems in Industry

In addition to being used in every department in a company, information systems are used in almost every industry or field in business. The airline industry develops Internet auction sites to offer discount fares and increase revenue. Investment firms use information systems to analyze stocks, bonds, options, the futures market and other financial instruments, and provide improved services to their customers. Banks use information systems to help make sound loans and good investments, as well as to provide online payment for account holders. The transportation industry uses information systems to schedule trucks and trains to deliver goods and services at the lowest cost. Publishing companies use information systems to analyze markets and to develop and publish newspapers, magazines and books. Private healthcare organizations use information systems to diagnose illnesses, plan medical treatment, track patient records and bill patients. Retail companies are using the web to take orders and provide customer service support. Retail companies also use information systems to help market products and services, manage inventory levels, control the supply chain and forecast demand. Power management and utility companies use information systems to monitor and control power generation and usage. Professional services firms employ information systems to improve the speed and quality of services they provide to customers. Management consulting firms use intranets and extranets to offer information on products, services, skill levels and past engagements to their consultants. These industries are discussed in more detail as we continue through the book.

1.5 Global Challenges in Information Systems

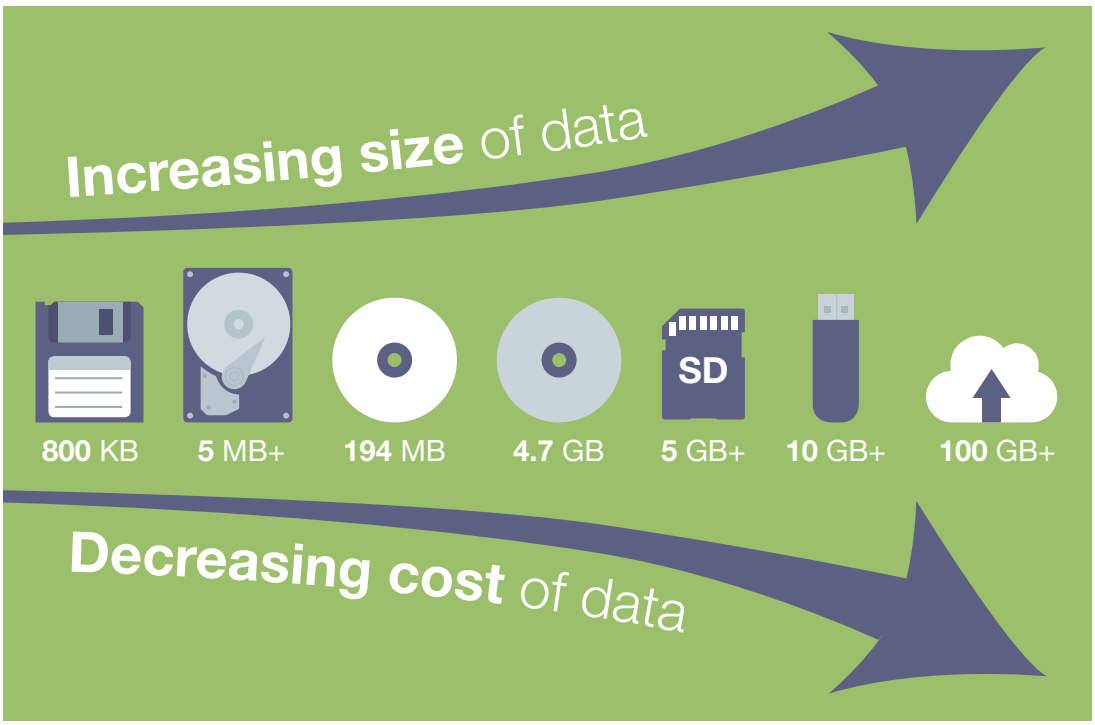
Changes in society as a result of increased international trade and cultural exchange, often called globalization, have always had a big impact on organizations and their information systems. In his book *The World Is Flat*, Thomas Friedman describes three eras of globalization (see Table 1.2).[20] According to Friedman, we have progressed from the globalization of countries to the globalization of multinational corporations and individuals. Today, people in remote areas can use the Internet to compete with, and contribute to, other people, the largest

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corporations and entire countries. These workers are empowered by high-speed Internet access, making the world seem smaller and effectively levelling the global playing field. In the Globalization 3 era, designing a new aeroplane or computer can be separated into smaller subtasks and then completed by a person or small group that can do the best job. These workers can be located in India, China, Russia, Europe and other areas of the world. The subtasks can then be combined or reassembled into the complete design. This approach can be used to prepare tax returns, diagnose a patient's medical condition, fix a broken computer and many other tasks.

Table 1.2 Eras of Globalization

Era	Dates	Characterized by
Globalization 1	Late 1400–1800	Countries with the power to explore and influence the world
Globalization 2	1800–2000	Multinational corporations that have plants, warehouses and offices around the world
Globalization 3	2000–today	Individuals from around the world who can compete and influence other people, corporations and countries by using the Internet and powerful technology tools



Increasing size of data: *As the cost of storing data has fallen, the world's need for data storage has increased.*

Today's information systems have led to greater globalization. High-speed Internet access and networks that can connect individuals and organizations around the world create more international opportunities. Global markets have expanded. People and companies can get products and services from around the world, instead of around the corner or across town.

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These opportunities, however, introduce numerous obstacles and issues, including challenges involving culture, language and many others.

- *Cultural challenges.* Countries and regional areas have their own cultures and customs that can significantly affect individuals and organizations involved in global trade.
- *Language challenges.* Language differences can make it difficult to translate exact meanings from one language to another.
- *Time and distance challenges.* Time and distance issues can be difficult to overcome for individuals and organizations involved with global trade in remote locations. Large time differences make it difficult to talk to people on the other side of the world. With long distance, it can take days to get a product, a critical part or a piece of equipment from one location to another location.
- *Infrastructure challenges.* High-quality electricity and water might not be available in certain parts of the world. Telephone services, Internet connections and skilled employees might be expensive or not readily available.
- *Currency challenges.* The value of different currencies can vary significantly over time, making international trade more difficult and complex.
- *Product and service challenges.* Traditional products that are physical or tangible, such as a car or bicycle, can be difficult to deliver to the global market. However, electronic products (e-products) and electronic services (e-services) can be delivered to customers electronically, over the phone, networks, through the Internet or other electronic means. Software, music, books, manuals, and help and advice can all be delivered over the Internet.
- *Technology transfer issues.* Most governments don't allow certain military-related equipment and systems to be sold to some countries. Also, some believe that foreign companies are stealing the intellectual property, trade secrets, copyrighted materials, and counterfeiting products and services.[21, 22]
- *National laws.* Every country has a set of laws that must be obeyed by citizens and organizations operating in the country. These laws can deal with a variety of issues, including trade secrets, patents, copyrights, protection of personal or financial data, privacy and much more. Laws restricting how data enters or exits a country are often called 'trans-border data-flow laws'. Keeping track of these laws and incorporating them into the procedures and computer systems of multinational and trans-national organizations can be very difficult and time consuming, requiring expert legal advice.
- *Trade agreements.* Countries often enter into trade agreements with each other. The EU has trade agreements among its members.[23] The North American Free Trade Agreement (NAFTA) and the Central American Free Trade Agreement (CAFTA) are other examples.[24] Others include the Australia–United States Free Trade Agreement and agreements between Bolivia and Mexico, Canada and Costa Rica, Canada and Israel, Chile and Korea, Mexico and Japan, the USA and Jordan and many others.[25]

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Summary

The value of information is directly linked to how it helps decision makers achieve the organizational goals. Information systems are used in almost every imaginable career area. Regardless of your chosen career, you will find that information systems are indispensable tools to help you achieve your goals. Learning about information systems can help you get your first job, earn promotions and advance your career.

Information is a collection of facts. To be valuable, information must have several characteristics: it should be accurate, complete, economical to produce, flexible, reliable, relevant, simple to understand, timely, verifiable, accessible and secure. The value of information is directly linked to how it helps people achieve their organization's goals.

Computers and information systems are constantly making it possible for organizations to improve the way they conduct business. A system is a set of elements that interact to accomplish a goal or set of objectives. The components of a system include inputs, processing mechanisms and outputs. A system uses feedback to monitor and control its operation to make sure that it continues to meet its goals and objectives.

System performance is measured by its efficiency and effectiveness. Efficiency is a measure of what is produced divided by what is consumed; effectiveness measures the extent to which a system achieves its goals. A system's performance standard is a specific objective.

Knowing the potential impact of information systems and having the ability to put this knowledge to work can result in a successful personal career, organizations that reach their goals and a society with a higher quality of life. Information systems are sets of interrelated elements that collect (input), manipulate and store (process), and disseminate (output) data and information. Input is the activity of capturing and gathering new data, processing involves converting or transforming data into useful outputs, and output involves producing useful information. Feedback is the output that is used to make adjustments or changes to input or processing activities.

The components of a computer-based information system (CBIS) include hardware, software, databases, telecommunications and the Internet, people and procedures. The types of CBISs that organizations use can be classified into: (1) e-commerce and m-commerce, TPS and ERP systems; (2) MIS and DSS; and (3) specialized business information systems. The key to understanding these types of systems begins with learning their fundamentals.

E-commerce involves any business transaction executed electronically between parties such as companies (business to business), companies and consumers (business to consumer), business and the public sector, and consumers and the public sector. The major volume of e-commerce and its fastest-growing segment is business-to-business transactions that make purchasing easier for big corporations. E-commerce also offers opportunities for small businesses to market and sell at a low cost worldwide, thus allowing them to enter the global market right from start-up. M-commerce involves 'anytime, anywhere' computing that relies on wireless networks and systems.

The most fundamental system is the transaction processing system (TPS). A transaction is any business-related exchange. The TPS handles the large volume of business transactions that occur daily within an organization. An enterprise resource planning (ERP) system is a set of integrated programs that can manage the vital business operations for an entire multisite, global organization. A management information system (MIS) uses the information from a TPS to generate information useful for management decision making.

A decision support system (DSS) is an organized collection of people, procedures, databases and devices that help make problem-specific decisions. A DSS differs from an MIS in the support given to users, the emphasis on decisions, the development and approach, and the system components, speed and output.

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Specialized business information systems include knowledge management, artificial intelligence, expert and virtual reality systems. Knowledge management systems are organized collections of people, procedures, software, databases and devices used to create, store, share and use the organization's knowledge and experience. Artificial intelligence (AI) includes a wide range of systems in which the computer takes on the characteristics of human intelligence. Robotics is an area of AI in which machines perform complex, dangerous, routine or boring tasks, such as welding car frames or assembling computer systems and components. Vision systems allow robots and other devices to have 'sight' and to store and process visual images. Natural language processing involves computers interpreting and acting on verbal or written commands in English, Spanish or other human languages. Learning systems let computers learn from past mistakes or experiences, such as playing games or making business decisions, while neural networks is a branch of AI that allows computers to recognize and act on patterns or trends. An expert system (ES) is designed to act as an expert consultant to a user who is seeking advice about a specific situation. Originally, the term 'virtual reality' referred to immersive virtual reality, in which the user becomes fully immersed in an artificial, computer-generated 3D world. Virtual reality can also refer to applications that are not fully immersive, such as mouse-controlled navigation through a 3D environment on a graphics monitor, stereo viewing from the monitor via stereo glasses, stereo projection systems and others.

System users, business managers and information systems professionals must work together to build a successful information system. Systems development involves creating or modifying existing business systems. The major steps of this process and their goals include systems investigation (gain a clear understanding of what the problem is), systems analysis (define what the system must do to solve the problem), systems design (determine exactly how the system will work to meet the business needs), systems implementation (create or acquire the various system components defined in the design step), and systems maintenance and review (maintain and then modify the system so that it continues to meet changing business needs).

Information systems must be applied thoughtfully and carefully so that society, business and industry can reap their enormous benefits. Information systems play a fundamental and ever-expanding role in society, business and industry. But their use can also raise serious security, privacy and ethical issues. Effective information systems can have a major impact on corporate strategy and organizational success. Businesses around the globe are enjoying better safety and service, greater efficiency and effectiveness, reduced expenses, and improved decision making and control because of information systems. Individuals who can help their businesses realize these benefits will be in demand well into the future.

Computer and information systems literacy are prerequisites for numerous job opportunities, and not only in the IS field. Computer literacy is knowledge of computer systems and equipment, and information systems literacy is knowledge of how data and information are used by individuals, groups and organizations. Today, information systems are used in all the functional areas of business, including accounting, finance, sales, marketing, manufacturing, human resource management and legal information systems. Information systems are also used in every industry, such as airlines, investment firms, banks, transportation companies, publishing companies, healthcare, retail, power management, professional services and more.

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Self-Assessment Test

1. Input is _____ to produce output.
2. A _____ is a specific objective of a system.
3. _____ is information from a system that is used to make changes to the input.
4. Providing software and data storage via a web browser is usually called _____.
5. The most important element in an information system is the _____.
6. An _____ supports long-term strategic decision making.
7. A business related exchange is known as a _____.
8. M-commerce involves paying for goods and services using a _____.
9. The activity of creating an information system is called _____.
10. When a person has the ability to use a computer they are known as being _____.

Review Questions

1. Describe what an information system is. What are its main elements?
2. Define the word 'system'.
3. What are the main components of a system? What does each of them do?
4. What are the characteristics of good quality information?
5. Describe what a TPS does.
6. What is m-commerce?
7. Define telecommunications.
8. What is virtual reality and how could it be used in an organization?
9. What are the main steps in systems development?
10. What are some of the global challenges in the field of information systems?

Discussion Questions

1. List the areas of your life that are impacted by information systems. Select a few of these and describe how things would be different without IS.
2. If driverless technology was perfected, how would you organize the country's daily commute to work? Would we all own a car or would you take a different approach?

Web Exercises

1. Search for information about a system that you interact with (it doesn't have to be a software system). Describe the main input, processing and output.
2. Find examples of software that allows working from home including perhaps during the coronavirus (COVID-19) pandemic. What are the features that each example offers that facilitate home working?

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1

Case One

Boaty McBoatface: The Online Survey That Went Viral

Oh dear. It has happened before and it will happen again. A serious organization naively assumes that the Internet-using public will take its survey seriously.

Having just spent over €300 million on a new Arctic-going research vessel, the Natural Environment Research Council (NERC) thought it would stir up interest in its work by running an online poll to decide what to name it. Perhaps the public would suggest naming it after a famous explorer. Or maybe Sir David Attenborough, a name synonymous in the UK with TV nature programmes. A few weeks later and the results were in. The winning suggestion? 'Boaty McBoatface'.

The suggestion received 124,109 votes, a massive four times as many as the runner up, Poppy-Mai, named after a 16-month-old girl with incurable cancer. The chief executive of the NERC, Duncan Wingham, was then faced with the dilemma of choosing between a credible name and overwhelming public opinion.

As mentioned, this is not the first time this sort of thing has happened. In 2012, soft drinks maker PepsiCo ran a survey to 'Dub the Dew' and give their new flavour a brand name. Unfortunately pranksters from the online community 4Chan stepped in and the name 'Hitler Did Nothing Wrong' rose to the top of the suggestions (although other communities were also blamed for this).

The UK government was quick to put the breaks on Boaty McBoatface. Jo Johnson, the then science minister said, 'The new royal research ship will be sailing into the world's iciest waters to address global challenges that affect the lives of hundreds of millions of people, including global warming, the melting of polar ice and rising sea levels. That's why we want a name that lasts longer than a social media news cycle and reflects the serious nature of the science it will be doing. There are many excellent suggestions among the 7,000 names put forward by members of the public and we'll make a decision as to which one should be put forward for the royal warrant when we've had a chance to review them all'.

There was even more to answer for. A parliamentary inquiry into the affair criticized NERC and its partners in the British government for not having sufficiently planned for what would happen after the naming contest ended. NERC had invited the public to engage with their new research ship but didn't specify what engagement they were looking for and how to proceed if and when people actually engaged. *The Atlantic* magazine asked 'What's the point of getting people involved if their involvement stops at voting in an online poll? It's a bit like asking someone on a date without gaming out what you'll do if you get a "yes"'.

Some people thought the NERC could compromise and use Boaty McBoatface in some way without actually giving that name to the ship (perhaps by using the name for a raft or life vessel).

In fact, NERC did indeed announce that Boaty McBoatface would be the name of the ship's remote submarine and the ship itself would be called *The Sir David Attenborough*.

Questions

1. If you were chief executive of the NERC, what would your decision be?
2. Are online surveys of this kind just too dangerous for an organization?
3. What are the advantages of this sort of survey?
4. Is there no such thing as bad publicity? Who would have heard of this €300m ship if it had not been for the Boaty McBoatface suggestion?

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Case Two

Health Information Systems in South Africa

Hospitals and health workers have long recognized the importance of good information. A knowledge of diagnoses made by doctors in the past, and a record of treatments that were given to patients along with the outcomes, all help to create the most promising treatment plans for future patients. Information on the geography of health problems also helps to create a national strategy that can be used to make health-related decisions and plan responses to disease outbreaks. The data may even be used to predict and prevent those outbreaks.

South Africa's Health Information Systems Program is a not-for-profit non-governmental organization that aims to empower healthcare workers and decision makers to improve the coverage, quality and efficiency of health services through the use of open standards and data exchange mechanisms. They subscribe to an open source philosophy, freely sharing their training materials and software solutions.

Their staff members include nine software developers and ten database administrators who create web-based and mobile solutions. This use of web standards means that their software will run on most operating systems and on most devices, which means there will be few, if any, access problems. One of their main systems, called DHIS2, is used to collect a range of data. This includes case-based data from in-patient admissions, essentially recording the who, what, when and where of patients being seen. Data is also stored on disease outbreaks and responses. New mothers and young children are seen as being particularly vulnerable, and so data is stored to improve their care. The results of lab tests and information on stocks of drugs and other treatments in hospitals are also recorded, as is data on patient deaths. The system includes a variety of ways to view all of this including charts and graphs. Geographic features that allow users to view maps of hospitals and to visualize the catchment areas for each are also available. All of this is seen through a clear, easy to navigate, information dashboard. The COVID-19 pandemic has also highlighted how important it is to have information and systems in place to deal with health crisis situations.

One important but previously often overlooked feature that is included in DHIS2 is social networking. In other words, DHIS2 allows for communication between users. This often encourages knowledge sharing and means that health professionals do not have to work in isolation. Patients too are realizing the importance of sharing their data. For instance, in a case unrelated to the HIS program, South African members of PatientsLikeMe recently contributed to an online survey of drug use which gave health professionals vital information. PatientsLikeMe is a web-based community of patients who compare the treatments they receive, and discuss their symptoms and experiences. The goal is for patients to support each other and to use the data generated to develop new treatment plans using cutting edge analytics. Towards that goal, in 2016 the NHS in the UK gave a company owned by Google access to 1.6 million patients from three hospitals. One of the goals is to develop software to alert staff to patients at risk of deterioration and death through kidney failure. However, local papers have reported that some patients are concerned about privacy and sharing sensitive data with a private company.

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Questions

1. What devices could be used to collect health information and what are the advantages and disadvantages of each?
2. What are the advantages of health professionals discussing patients with each other online? Are there any disadvantages?
3. Why do you think the Health Information Systems Program uses an open source philosophy?
4. Why are patients concerned with sharing sensitive data with a private company?

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Case Three

Australian Drones Join The Postal Service

Australia's national postal service occassionally uses drones to make deliveries. Testing has begun that may in the near future see some customers receiving parcels from the sky. This could be good news for those who live in more remote areas and for people whose mailboxes are a distance from their doors, such as farmers, some of whom have to drive many miles to the gates of their property where deliveries are usually left.

The drones are remote controlled by an operator and have been built by Australian company ARI Labs. They feature a parachute in case of engine failure and have a camera at the front giving the operator a real-time view of what is ahead of the drone. The customer lays out a landing grid that the operator can see (essentially a brightly coloured picnic blanket) when they get close to the delivery address.

Some operators use a headset to immerse themselves in the experience of flying. This involves wearing goggles that display the feed from the camera and give them the sense that they are actually on board the drone. Doing this often makes people physically sick when they start to use it, and even after that initial period, some people will never enjoy the sensation. It's like sea sickness and can last for several hours. There are several explanations for it but the most likely seems to be that the body is getting signals through the eyes that it is moving, but nothing from the rest of the senses.

When the drone is nearing its destination, a message is sent to the receiver that it is about to land and to put out the landing grid. What happens next is being tested, but it's likely to be that the drone lands, drops the parcel on the ground and then takes off again, allowing the receiver to then go safely in and pick up their package, having never interacted directly with the drone.

Others are interested in this technology. Drones have been used to deliver contraband into prisons and to smuggle drugs across borders. In the UK, there were 33 incidents of smuggling contraband into prisons involving devices in 2015. This has caused some police forces to train up eagles to grab them in mid flight!

Australia's Civil Aviation Safety Authority is monitoring the tests carefully and will decide whether to allow the postal service to go ahead with the scheme. Australia Post CEO Ahmed Favour said that although the company hoped to get the drones delivering soon, they would only start shipping parcels 'once we are 100 percent sure that it's safe and reliable'. As yet, no details have been released about the cost of the service to customers.

Questions

1. What precautions would have to be taken to get this technology off the ground?
2. What other uses can you think of for this technology?

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3 What alternatives are there to the use of a headset for drone operators?

4 Will all police forces need to start training eagles?

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